

Question	Answer	Marks
1(a)	$v_x = (6.0^2 - 4.8^2)^{1/2} = 3.6 \text{ (ms}^{-1}\text{)}$ or $6.0 \sin \theta = 4.8 \text{ (so } \theta = 53.1^\circ \text{) and } v_x = 6.0 \cos 53.1^\circ = 3.6 \text{ (ms}^{-1}\text{)}$	A1
1(b)(i)	straight line from (0, 4.8) to (0.49, 0)	M1
	straight line continues with same slope to (0.98, -4.8) (labelled Y)	A1
1(b)(ii)	a horizontal line	M1
	from (0, 3.6) to (0.98, 3.6) (labelled X)	A1
1(c)	$s = ut + \frac{1}{2}at^2$ $= (4.8 \times 0.49) + (\frac{1}{2} \times -9.81 \times 0.49^2)$ or $s = \frac{1}{2}(u + v)t$ or area under graph $= \frac{1}{2} \times (4.8 + 0) \times 0.49$ or $s = vt - \frac{1}{2}at^2$ $= \frac{1}{2} \times 9.81 \times 0.49^2$ or $v^2 = u^2 + 2as$ $s = 4.8^2 / (2 \times 9.81)$	C1
	$s = 1.2 \text{ m}$	A1

Question	Answer	Marks
1(d)	$(\Delta)E = mg(\Delta)h$	C1
	$E = \frac{1}{2}mv^2$	C1
	ratio = $(\frac{1}{2} \times m \times 3.6^2) / (m \times 9.81 \times 1.2)$ or ratio = $[(\frac{1}{2} \times m \times 6.0^2) - (m \times 9.81 \times 1.2)] / (m \times 9.81 \times 1.2)$ or ratio = $(\frac{1}{2} \times m \times 3.6^2) / (\frac{1}{2} \times m \times 4.8^2)$	C1
	ratio = 0.56	A1
1(e)	(force due to) air resistance acts in opposite direction to the velocity or (with air resistance, average) resultant force is larger (than weight)	B1

Question	Answer	Marks
2(a)	ampere	B2